

Assessing the Responsiveness of U.S. Democracy, Human Rights, and Governance Foreign Aid to Recipient Country Democracy Level (2001-2024)

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Python Programming II - 90-819
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1. Introduction

How did United States (U.S.) Democracy, Human Rights, and Governance (DRG) foreign assistance respond to democratic backsliding or improvements in recipient countries between 2001 and 2024? I compared aid obligations with independent democracy measures to examine how funding relates to democratic shifts. Against a backdrop of U.S. Agency for International Development (USAID) cuts and program rollbacks, this analysis provides historical context for debates about democracy assistance and taxpayer accountability, examining whether U.S. commitments to democratic values align with funding patterns.

2. Data, Cleaning, and Merging

I combined the Varieties of Democracy (V-Dem) country-year data with [ForeignAssistance.gov](https://www.foreignassistance.gov) data for 2001-2024, capturing the post-9/11 period before 2025 funding disruptions.^{1,2}

V-Dem: I retained the electoral democracy, liberal democracy, civil society participation, and regime classification columns, adding one-year lags to examine subsequent funding responses. I filled in three missing liberal-democracy scores based on adjacent scores, assuming minimal changes in the short term.

Foreign Assistance: I aggregated DRG obligations—legally binding aid commitments—by country and fiscal year, using inflation-adjusted millions of dollars for comparability over time. I excluded seven small legacy Czechoslovakia records because the dissolved state lacked a country code; they likely reflected belated accounting adjustments.

I outer-joined the datasets using country code and year, acknowledging the mismatch between calendar years and October-September U.S. fiscal years. I assigned missing aid values zero, assuming absent records represented no obligations rather than unreported

¹ Michael Coppedge et al., “V-Dem Country-Year Dataset v16” (Varieties of Democracy [V-Dem] Project, 2026), <https://doi.org/10.23696/vdemds26>.

² U.S. Agency for International Development and U.S. Department of State, *ForeignAssistance.gov*, dataset, 2026, accessed October 9, 2026, <https://www.foreignassistance.gov/data>.

funding. My repository documents variable definitions, cleaning procedures, and reproducible code.

3. Exploratory Findings & Analysis

I explored variable distributions using descriptive statistics and histograms. Table 1 summarizes 4,936 aid observations: mean DRG aid (\$20.96 million) exceeds its median (\$1.43 million), revealing right-skewness; negative obligations may reflect accounting adjustments. Across 4,280 observations, mean civil society participation (0.66) exceeds electoral democracy (0.51) and liberal democracy (0.40).

	N	Mean	Std Dev	Min	25th Pct	Median	75th Pct	Max
aid_millions	4936	20.96	146.19	-60.37	0.00	1.43	11.38	6173.09
v2x_libdem	4280	0.40	0.27	0.01	0.15	0.37	0.64	0.90
v2x_cspart	4280	0.66	0.24	0.02	0.53	0.72	0.84	0.99
v2x_polyarchy	4280	0.51	0.26	0.01	0.28	0.51	0.76	0.92

Table 1. Descriptive statistics of numerical variables

Figure 1 confirms strong right-skewness, with many aid observations below \$2 million. I therefore log-transformed aid to reduce extreme values' influence.

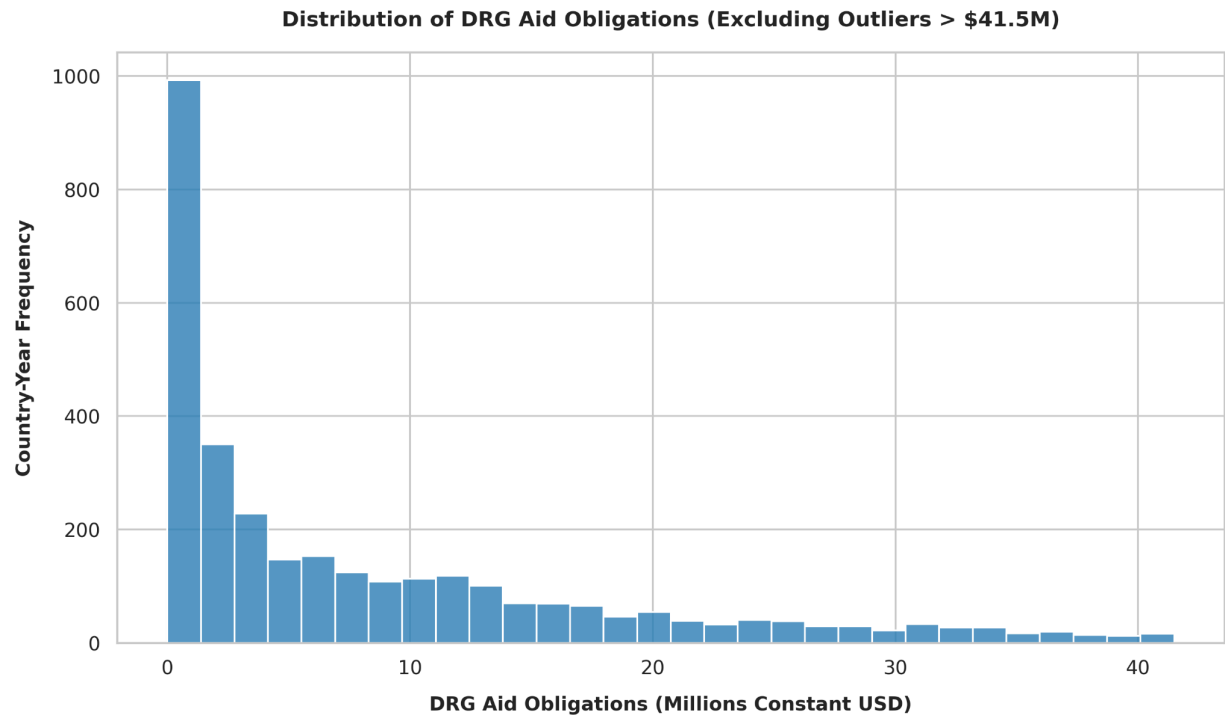


Figure 1. Histogram of DRG aid obligations

Figure 2 shows roughly bell-shaped logged aid, peaking near 1, and moderately left-skewed civil society participation. Electoral and liberal democracy have less distinctive patterns, with peaks near 0.1 and 0.9, respectively. High liberal-democracy scores remain uncommon.

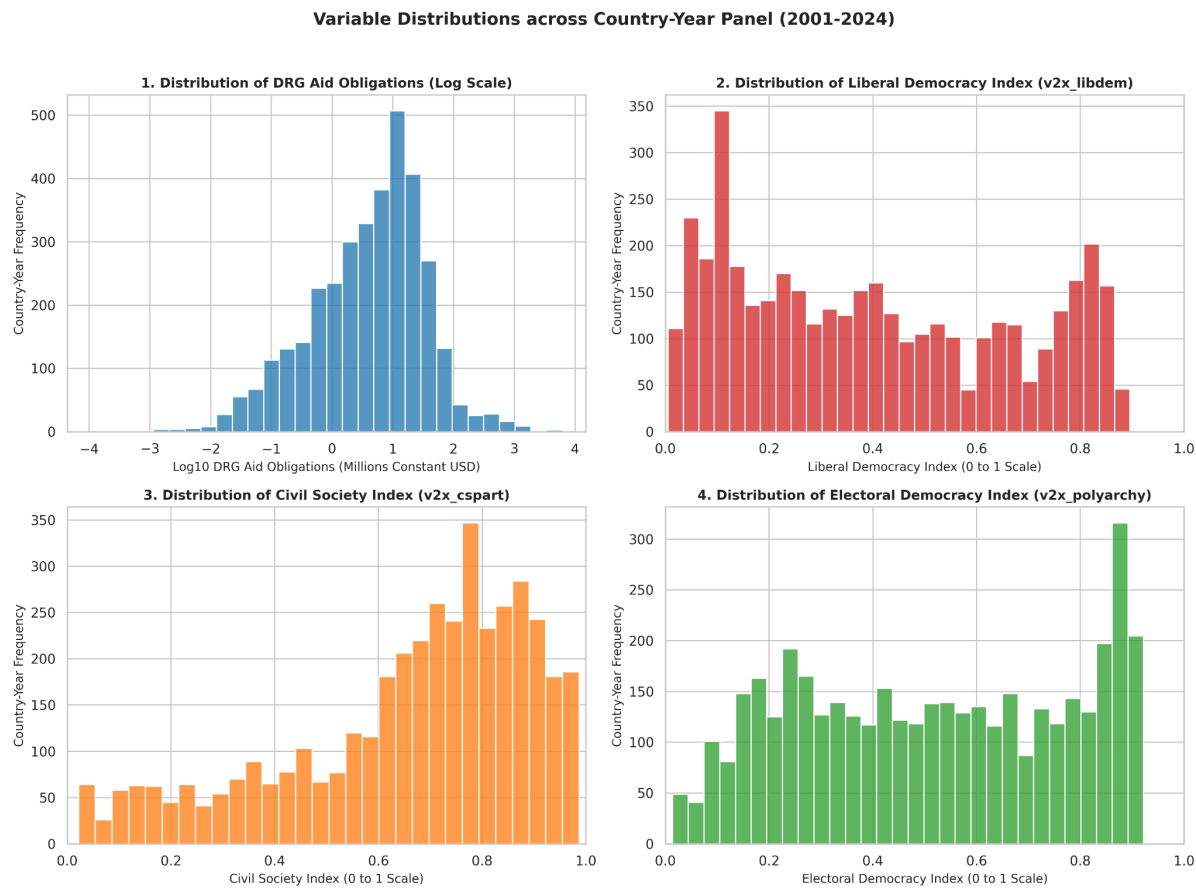


Figure 2. Histograms of log aid and democracy indices

Figure 3 overlays a bar plot of total global annual DRG aid with global average democracy indices by year. Aid fluctuates from approximately \$2-8 billion per year, with a slight downward trend. All democracy indices peaked in 2012, then declined, hitting their time-period minimums by 2024. There is no clear relationship between global annual DRG aid and global average democracy.

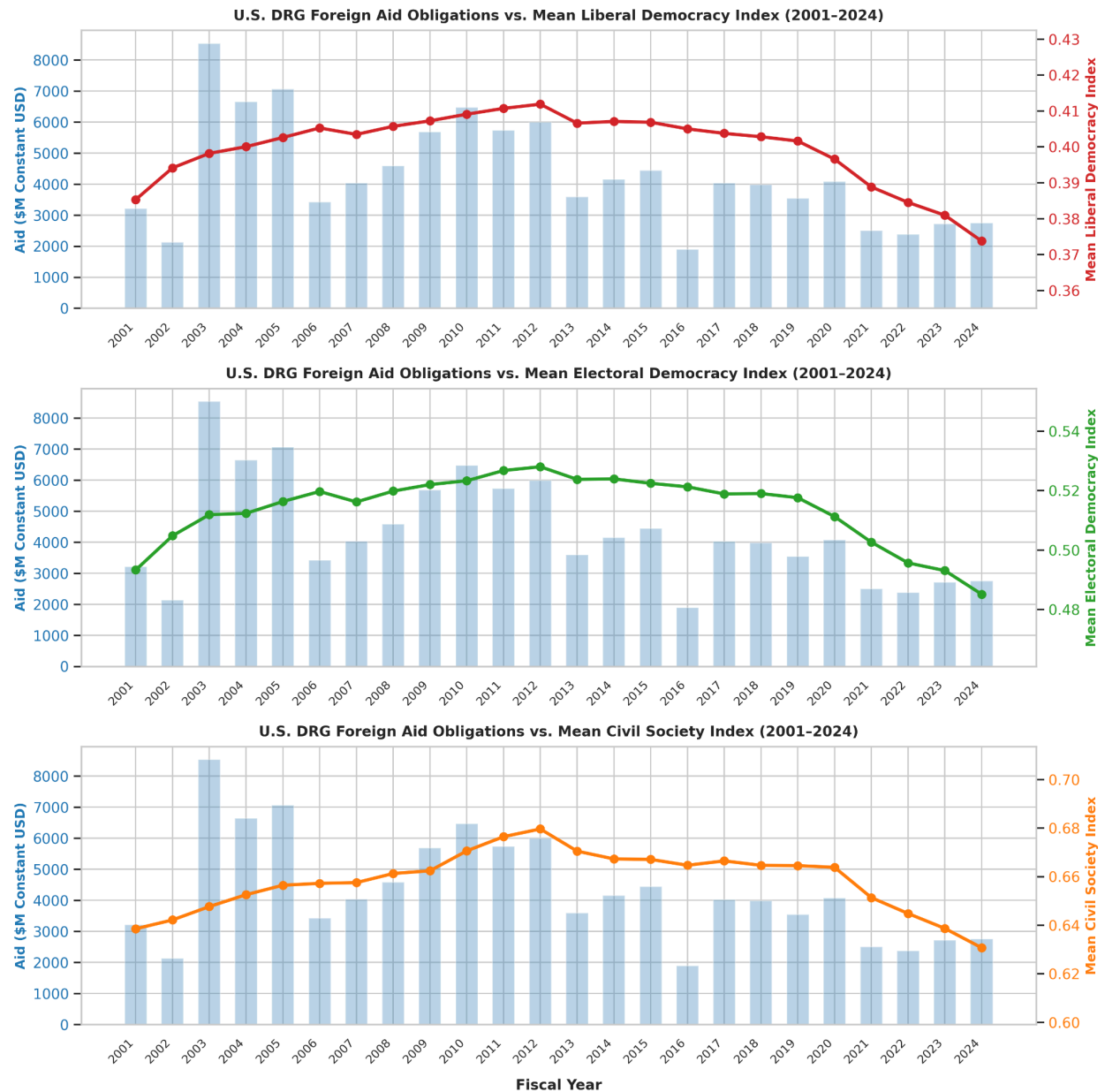


Figure 3. Bar plot of global DRG aid and line plots of global average democracy indices by year

Figure 4 shows strong positive correlations among democracy indices, moderate positive correlations between aid and its log, and weak-to-moderate negative correlations between log aid and democracy scores. Other correlations are weak. Nonlinear relationships may be obscured.

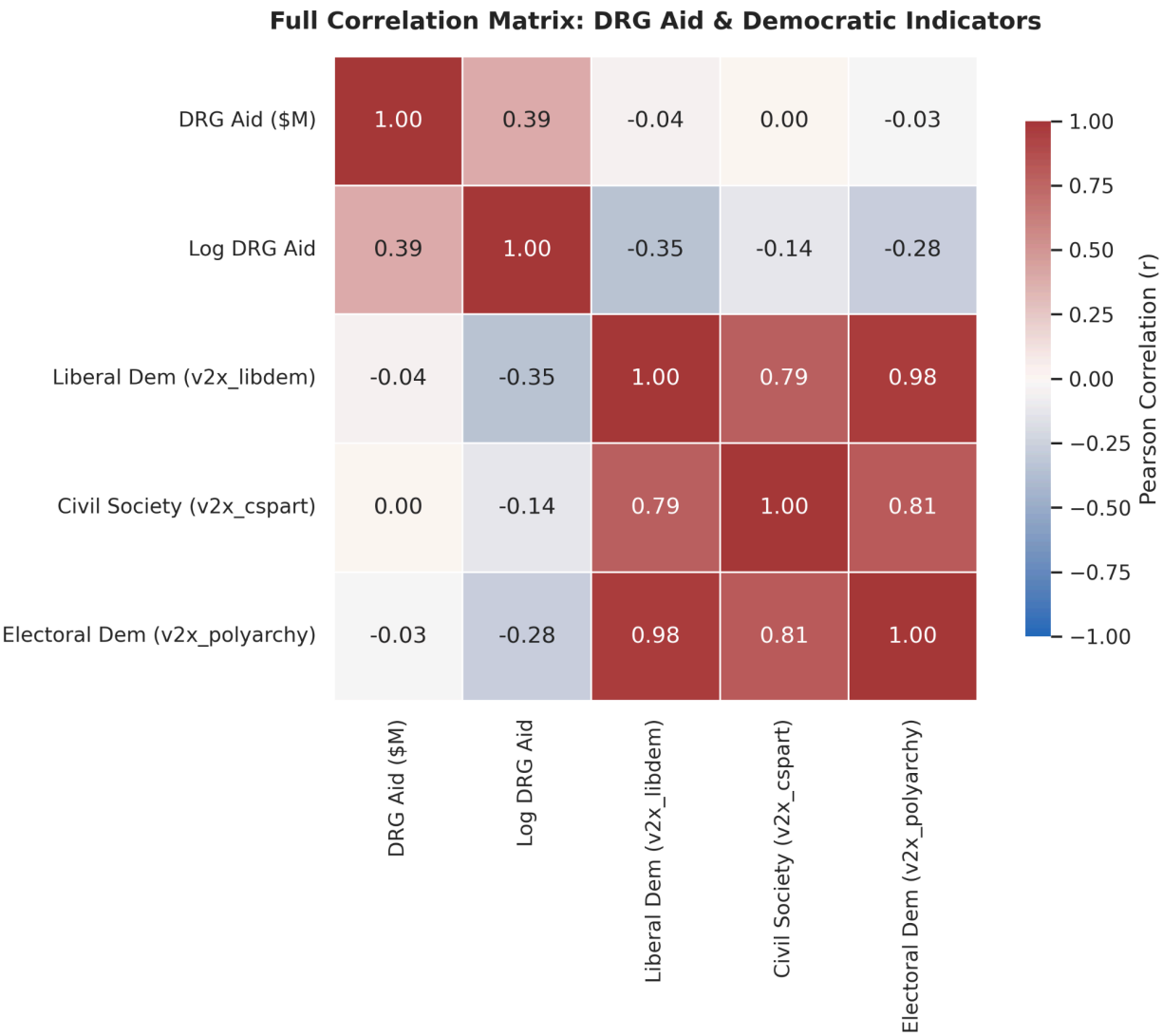


Figure 4. Correlation heatmap of numeric variables

Figures 5 through 7 reveal a contrasting within-country pattern: above average democracy scores correlate positively with next-year logged aid deviations. Slopes are 0.816 for liberal democracy, 0.676 for electoral democracy, and 0.991 for civil society; respective R² values are 0.0263, 0.0237, and 0.0636. Civil society shows the strongest association. However, all associations, although statistically significant, explain little variation and do not establish causality or policymakers’ motivations.

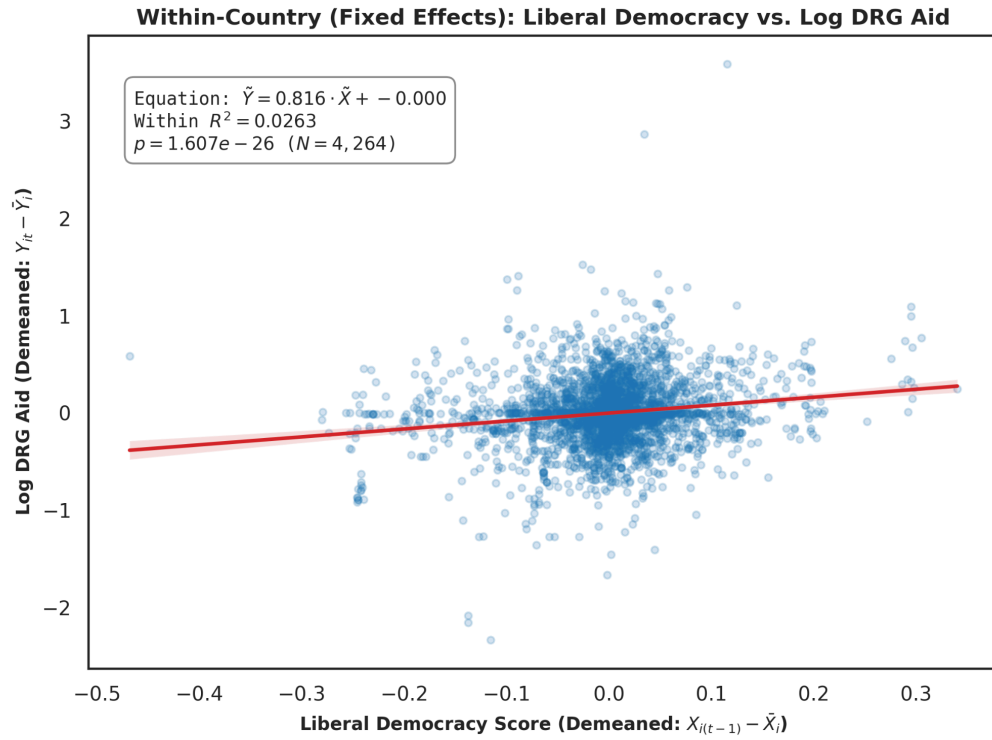


Figure 5. Scatter plot of demeaned lagged liberal democracy score and log DRG aid

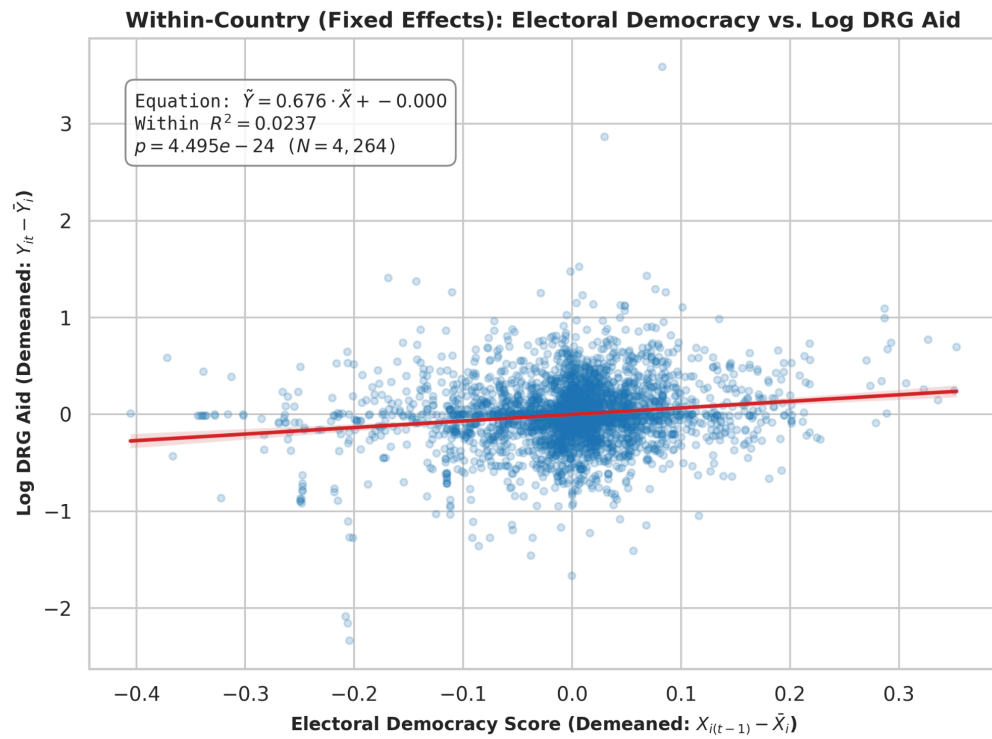


Figure 6. Scatter plot of demeaned lagged electoral democracy score and log DRG aid

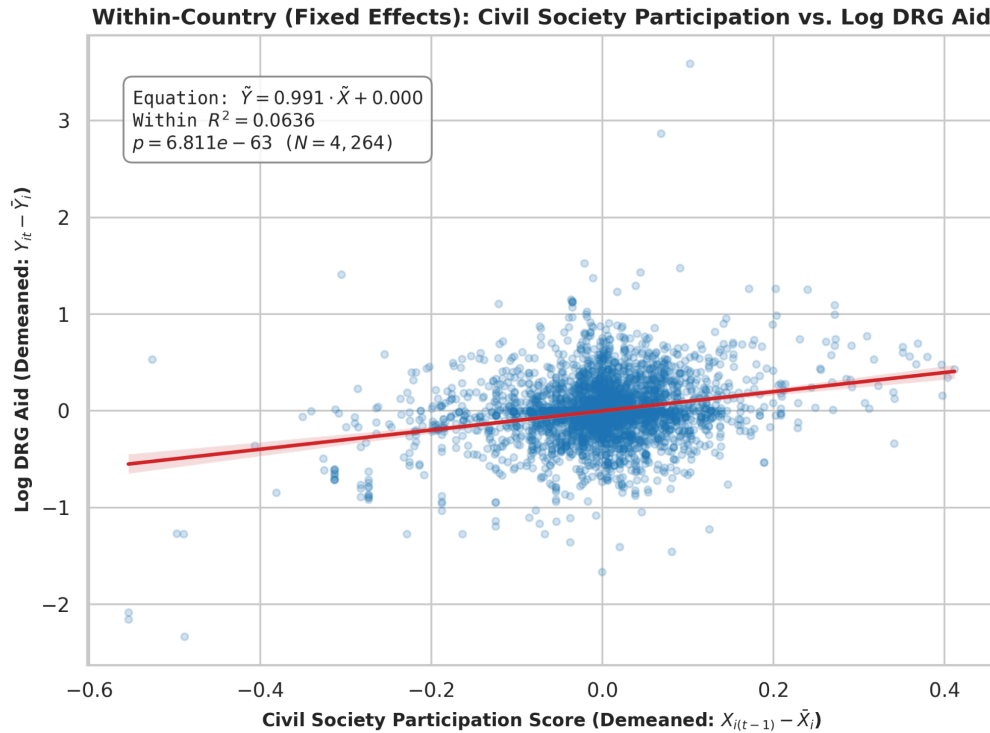


Figure 7. Scatter plot of demeaned lagged civil society participation score and log DRG aid

Figure 8 compares median country-year DRG aid to the previous year's regime type, using medians to limit outlier influence and 95% confidence intervals to show uncertainty. Electoral autocracies and electoral democracies receive the highest median aid; closed autocracies receive little, and liberal democracies virtually none.

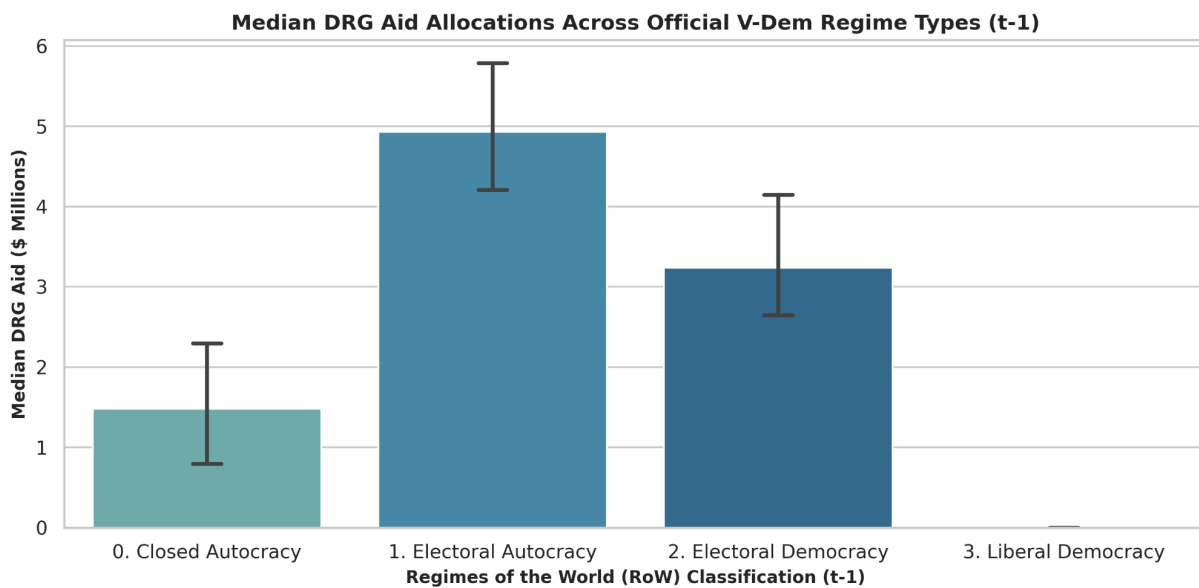


Figure 8. Bar plot of median DRG aid by regime type 1 year before

Figure 9 similarly shows aid concentrated in the middle regime type categories, with exceptions driven by Poland in 2003 and Iraq in 2004-2005. These unusually large obligations warrant further research to understand their context.

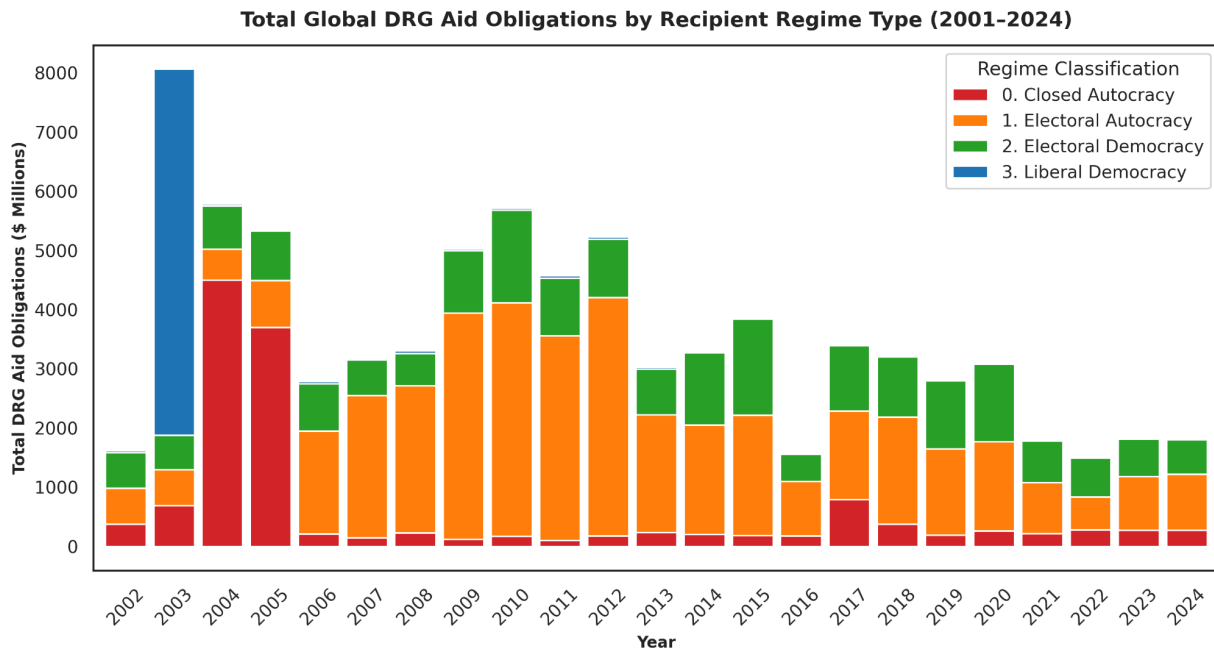


Figure 9. Stacked bar plot of DRG aid by regime type 1 year before over time

4. Policy Recommendations

My descriptive findings identify associations, not causal effects, and cannot clearly justify funding increases, reductions or program cuts. Aid concentrates in electoral autocracies and electoral democracies, suggesting that DRG funding tends to go to flawed regimes with potential openness to change and room for improvement, and weakly correlates with democracy indices.

I recommend country-level research because global averages may conceal divergent country trajectories. Future analysis should analyze aid per capita, account for recipient country income and local cost, and explore relationships with V-Dem's Episodes of Regime Transformation data. Testing lagged relationships in both directions could distinguish whether democratic changes precede funding adjustments or assistance precedes democratic changes.

For policymakers, I recommend transparent allocation criteria and program evaluations, rather than strictly tying aid amounts to democracy scores. Evaluations should measure outcomes, costs, and implementation constraints, and importantly, evaluations should be retained and published so implementers and policymakers can learn from them. My findings support investigating when DRG assistance works—not concluding that either expanding or eliminating it will improve democratic outcomes.